VS CODE · REMOTE SSH · QT

Debug a Qt GUI app on a remote Ubuntu PC

VS Code runs on Windows. The app runs on Ubuntu.
Its window shows on Windows.



OVERVIEW

How it works

YOU WORK HERE

Windows PC

VS Code

VcXsrv shows the window



THE APP RUNS HERE

Ubuntu PC

Source code and build

gdb and your app

Breakpoints work as usual. Only the window appears on Windows.

STEP 1

One-time setup

Windows PC

1. Create an SSH key and copy it to Ubuntu
2. Install VcXsrv and run XLaunch with display 0
3. Run setx DISPLAY "localhost:0.0"
4. Add ForwardX11 yes to the SSH config

Ubuntu PC

1. Install openssh-server, gdb, cmake, xauth
2. Install the Qt dev packages
3. Set X11Forwarding yes in sshd_config
4. Restart the ssh service

STEP 2

Connect and check

1

Connect

Press F1, then choose Remote-SSH: Connect to Host.

2

Install

Install C/C++ and CMake Tools on the remote.

3

Check

In the terminal, run:

```
echo $DISPLAY
```

```
xclock
```

Move on only when the clock opens on your Windows desktop.

STEP 3

Run and debug

1. Build with Debug symbols
2. Copy your DISPLAY value into launch.json
3. Set a breakpoint
4. Press F5

KEY SETTINGS IN LAUNCH.JSON

```
"type": "cppdbg",
"MIMode": "gdb",
"preLaunchTask": "build-debug",
DISPLAY=localhost:10.0
```

The window opens on Windows. A breakpoint pauses the app, so the window may look frozen.

TROUBLESHOOTING

If the window does not open

Symptom	Fix
echo $DISPLAY is empty	Turn on X11 forwarding, install xauth, reconnect, open a new terminal
xclock cannot open the window	Run setx DISPLAY "localhost:0.0" again, set XLaunch to display 0, restart VS Code
Authentication error	Tick Disable access control in XLaunch
Connection refused	Allow vcxsrv.exe in Windows Firewall

Our fix: match DISPLAY localhost:0.0 with XLaunch display 0.